

Project Proposal

Animal Crossing: New Horizons is a Nintendo Switch game where a player's customizable character moves to a deserted island and is tasked to develop the community through daily tasks, crafting, fishing, and bonding with recruited animal villagers. Animal Crossing: New Horizons offers a vast catalog of tens of thousands of decorative items such as furniture, decorations, insects, fish, and outfits that can be used to design a player's island. To build a community on the island, gifts are often exchanged between the players and villagers. This project aims to implement a machine-learning recommender system that recommends island items based on player's inventory, selected items, or the preferences of residing villagers. The recommended items will be similar in style, theme, or color.

The primary dataset this project will utilize is the Animal Crossing: New Horizons Data Spreadsheet, an extensive archive of all collectables, events, and villagers in the game that is regularly updated and maintained by the ACNH community. The data will be stored in a JSON file which will be easier to unpack and analyze in Python. The secondary dataset is from the Nookipedia API with an authorized API key. Most of the variables are categorical and are based on the characteristics of each item: variation, body title, pattern, DIY (whether the player can build item with resources), customizable (can change color of item with resources), dimensions, colors, exchange currency, source, concept, series, interactive, etc. There are up to 42 total attributes for each item, depending on the type of item, the number of attributes vary, and the categories of each item is separated by page in the spreadsheet. The numeric variables are internal ID, buying price, and selling price. There are in total ~18,000 items and 417 villagers with 29 attributes. The recommender system will take a collection of target items and find the nearest neighboring items as well as items that align with the preferences of a villager.

Since recommendations are reliant on the similarity of the nearest neighboring items in the data, all categorical variables will need to be properly encoded or normalized. NA values may need to be kept as well. The most efficient models to calculate and promote these recommendations are K-means clustering/unsupervised learning, nearest neighbor recommendation, item based collaborative filtering, and content-based filtering. Content-based filtering prevents the issue of "cold-start", a common recommender system challenge. This project can enable players to tailor their playing experience to their goals to earn what is needed for items and build their desired aesthetic. A recommender system for island items fosters creativity and a more personalized experience while making a sandbox open-world game continuously engaging and entertaining.

References

Nookipedia. (n.d.). *Nookipedia API*. Retrieved February 12, 2026, from <https://nookipedia.com/api>

ACNH Community. (n.d.). *Animal Crossing: New Horizons Data Spreadsheet* [Data set]. Retrieved February 12, 2026, from <https://docs.google.com/spreadsheets/d/e/2PACX-1vTGrlfAl5ybCvailux5kEbermRFZe6aooAs7l1iVrJF27DrXSOJQxxEcQXzlw6KRacx1721da2oN2SM/pubhtml>